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Collective Self-Esteem and Ingroup Bias

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In addition to personal self-esteem, we propose that there is a second type of self-esteem, collective self-esteem. People who are high in trait collective self-esteem should be more likely to react to threats to collective self-esteem by derogating outgroups and enhancing the ingroup. In a study using the minimal intergroup paradigm, trait personal and collective self-esteem were measured, and subjects received information about the average performance of their group. Subjects high in collective self-esteem varied their ratings of above-average and below-average scorers on the test in an ingroup-enhancing fashion, whereas those low in collective self-esteem did not. Analyses based on personal self-esteem did not show this interaction. We conclude that collective self-esteem is an individual difference variable that may moderate the attempt to maintain a positive social identity. The relation between collective and personal self-esteem is discussed.

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According to social identity theory (e.g., Tajfel, 1982; Tajfel & Turner, 1979, 1986; Turner, 1982), the self-concept has two distinct aspects. One is personal identity, which includes beliefs about one's skills, abilities, or attributes such as attractiveness or intelligence. The second is social identity, (what we call collective identity), defined as "that aspect of the individuals' self-concept which derives from their knowledge of their membership in a social group (or groups) together with the value and emotional significance attached to that membership" (Tajfel, 1981, p. 255).[1] Whereas personal identity concerns one's individual characteristics, social or collective identity concerns the characteristics of one's groups, which may or may not also characterize oneself as an individual.

Consistent with several theoretical perspectives on the self-concept (e.g., Greenwald, 1980; Rosenberg, 1979; Taylor & Brown, 1988; Tesser & Campbell, 1983; Wills, 1981), social identity theory posits that individuals are motivated to achieve or maintain a high level of self-esteem. However, whereas other perspectives focus on maintaining a positive personal identity (i.e., personal self-esteem), social identity theory is primarily concerned with the motivation to maintain a positive social identity (i.e., collective self-esteem). The theory proposes that when confronted with a threat to their social identity, people maintain a positive social identity by identifying or creating favorable comparisons between their own group(s) and outgroup(s). Thus, individuals will discriminate against or derogate outgroup members relative to the ingroup to create favorable comparisons between the ingroup and outgroup. These comparisons result in a positive social identity, or high collective self-esteem.

Empirical research, typically using the minimal intergroup paradigm, in which individuals are divided into groups on the basis of trivial or even arbitrary distinctions, is generally consistent with these predictions. Even when the self is explicitly excluded from evaluations, and even when the self does not benefit from the rewards, people will evaluate ingroup members more favorably than outgroup members and allocate more rewards to members of the ingroup than to the outgroup (see Brewer, 1979, for a review). Thus, even under these minimal intergroup conditions, in which group boundaries are essentially meaningless, and in the absence of any history of hostility, rivalry, or conflict between groups, people show bias in favor of the ingroup. Social identity theory provides a compelling account of prejudice and discrimination against outgroups by linking such behavior to the maintenance of a positive self-concept, specifically a positive social identity.

Current versions of social identity theory have not been concerned with individual differences in the extent to which people successfully achieve a positive social identity. However, a great deal of research on personal identity has indicated that people vary in the extent to which they possess a positive personal identity (i.e., high personal self-esteem). A large number of scales that measure personal self-esteem have been developed and empirically validated (see Wylie, 1974, for a review).

Although social identity theory has not directly addressed this issue, we have proposed that there are stable individual differences in the extent to which people possess a positive social identity (i.e., high collective self-esteem) as well as a positive personal identity. Indeed, Luhtanen and Crocker (1989, in press) have recently developed a measure of trait collective self-esteem that assesses the extent to which individuals *generally*

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[1] We prefer the term *collective* rather than *social* identity to avoid confusion with other social aspects of the self-concept, such as the self in interpersonal (rather than intergroup) relationships. Jonathan Cheek, for example, has developed an unpublished measure of social identity that is concerned with this interpersonal, rather than intergroup, aspect of the self-concept.

evaluate their social groups positively.[2] As measured by this scale, trait collective self-esteem is empirically distinct from trait personal self-esteem.

We propose that level of trait collective self-esteem may moderate the extent to which individuals attempt to protect or enhance their social identity when it is threatened. Our predictions are based on research on personal self-esteem. A large number of studies using a variety of methods have shown that people who are high in personal self-esteem show a variety of self-enhancing or self-serving illusions, biases, or distortions, particularly when faced with a real or imagined threat to the personal self-concept (see Taylor & Brown, 1988, for a review). For example, people with high self-esteem are more likely to take credit for success and avoid blame for failure than are those low in self-esteem (cf. Tennen & Herzberger, 1987).

In one study, Crocker, Thompson, McGraw, and Ingerman (1987) found that subjects high in personal self-esteem responded to information about their personal performance on a test in a manner that indirectly enhanced the self, although they did not directly enhance the self. Specifically, on the relatively direct ratings of the self, subjects with both high and low self-esteem who were told they failed the test rated the self more negatively than those who were told they succeeded. However, on the relatively indirect ratings of above- and below-average scorers, high but not low self-esteem subjects who received success feedback rated above-average scorers on the test much more positively than below-average scorers; those who received failure feedback rated above-average scorers no more positively than below-average scorers. Thus, high self-esteem subjects indirectly enhanced the implications of their success and minimized the implications of their failure. Low self-esteem subjects, in contrast, did not alter their ratings of above- or below-average scorers on the test as a consequence of their own performance—they consistently rated above-average scorers more positively than below-average scorers. Thus, low self-esteem subjects did not show the self-enhancing distortions that characterized high self-esteem subjects.

Luhtanen and Crocker (1989, in press) have proposed that collective self-esteem might operate in group contexts in a manner analogous to personal self-esteem in individual contexts. Whereas personal self-esteem may moderate the extent to which one responds to a personal failure experience, collective self-esteem may moderate the extent to which one responds to a group failure experience. High collective self-esteem people may engage in *ingroup*-enhancing biases or distortions when faced with a threat to their collective identity, whereas low collective self-esteem subjects may show the absence of such ingroup-enhancing biases. Thus, the predictions of social identity theory may apply to high collective self-esteem individuals but not to those low in collective self-esteem.

To test this hypothesis, we conducted a study in which subjects high or low in collective self-esteem received *group* success or failure feedback, and then rated above- and below-average scorers on the test. Thus, the study was similar in method to the study by Crocker et al. (1987) described above, with the exceptions that trait collective self-esteem was measured instead of trait personal self-esteem, and social identity was threatened (via group performance feedback) instead of personal identity (via individual performance feedback). This study represents the first attempt to assess whether scores on the trait collective self-esteem scale moderate reactions to threats to social identity. As such, it is the first study to assess experimentally the validity of the collective self-esteem scale. If collective self-esteem predicts behavior in group contexts as personal self-esteem predicts behavior in individual contexts, then we predict that relative to low collective self-esteem subjects, those high in collective self-esteem will be more likely to enhance the ingroup indirectly by altering their evaluations of above- and below-average scorers, but will not be more likely to enhance the ingroup directly.

Method

Subjects

Subjects were 85 introductory psychology students who received partial course credit for their participation. They were run in groups of 6 to 10. Three subjects' data were discarded because of suspicion regarding the deception in the study (see later discussion), leaving a total of 82 subjects.

Procedure

Subjects were told that the purpose of the study was to investigate self-perceptions and average group performances on the Social–Cognitive Aptitude Test (SCAT; Crocker et al., 1987) and how they relate to interpersonal judgments and behaviors. They were told not to communicate with one another during the experiment, until asked to do so. They were then arbitrarily divided into two groups, following a procedure frequently used within the minimal intergroup paradigm (cf. Crocker & Schwartz, 1985; Crocker et al., 1987; Locksley, Ortiz, & Hepburn, 1980). Each subject drew from a box a slip of paper on which was written the letter A or B, indicating assignment to Group A or Group B. Subjects were aware of their own group assignment during the experiment, but not the group assignment of the other subjects present. To control the amount of time subjects expected to interact with ingroup and outgroup members, they were informed that there would be a brief intergroup exercise at the end of the study, in which they would have a chance to interact with all of the other subjects in the session. They were led to expect that they would interact with members of their own group and the other group for equal amounts of time (cf. Locksley et al., 1980).

Subjects were then given a booklet containing the Rosenberg (1965) self-esteem inventory (a measure of personal self-esteem) and a measure of collective self-esteem (Luhtanen & Crocker, 1989, in press). Subjects were told that all measures used in the study were to be anonymous, and only to write their group assignments and subject numbers on each measure.

The Rosenberg scale is a measure of global personal evaluations of the self, and includes such items as "I often feel like a failure" and "I feel I am a person of worth, at least on an equal basis with others." It is a well-validated measure of global personal self-esteem, with high internal consistency and test–retest reliability (Rosenberg, 1965; see also Wy-

[2] Breckler and his colleagues (Breckler, Greenwald, & Wiggins, 1986) have developed an unpublished measure that they label a collective self-esteem scale. However, the content of their scale is different from what we call collective self-esteem. Specifically, their scale measures personal self-esteem in a group context (e.g., "I am a good representative of my school"), rather than esteem for one's social groups. The Luhtanen and Crocker scale includes a subscale with this type of item. As predicted, this "membership" subscale was the subscale most strongly correlated with personal self-esteem.

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lie, 1974). It has been used in previous studies of self-esteem and ingroup bias (Crocker & Schwartz, 1985; Crocker et al., 1987).

The collective self-esteem scale consists of four subscales that assess various aspects of collective self-esteem. In addition to the scale measuring private collective self-esteem (i.e., the extent to which one evaluates one's social groups positively), which is the central focus of this study, the scale contains three additional subscales measuring other aspects of social identity, including (a) membership esteem (evaluations of oneself as a good member of the social groups to which one belongs), (b) public collective self-esteem (how others evaluate one's social groups), and (c) importance to identity (how important one's memberships in the social groups are to one's self-concept). Factor analyses support the distinctiveness of each of these four-item subscales. The collective self-esteem scale, and each of its subscales, has high internal consistency (alphas > .83) and acceptable test–retest reliability (r = .68 for the total scale, r = .62 for the private collective self-esteem subscale) over a 6-week interval. It is moderately correlated with the Rosenberg self-esteem scale in this (r = .34) and other samples, and with other measures of personal self-esteem (Luhtanen & Crocker, 1989, in press). Unlike measures of personal self-esteem, however, it is uncorrelated with the Marlowe–Crown measure of social desirability (Crowne & Marlowe, 1964). More complete information about the development of the scale and its psychometric properties can be found in Luhtanen and Crocker (1989, in press).

The private collective self-esteem subscale measures global evaluations of one's ascribed social groups, and is the group-level equivalent of global personal self-esteem. Hence, scores on this subscale were expected to interact with group threats in a manner parallel to interactions between personal self-esteem and personal threat.[3] The subscale contains the following four items: "I feel good about the social groups I belong to"; "In general, I'm glad to be a member of the social groups I belong to"; "Overall, I often feel that the social groups of which I am a member are not worthwhile"; and "I often regret that I belong to some of the social groups I do" (the last two items are reverse-scored). The instructions stated:

> We are all members of different social groups or categories. Some of these social groups or categories pertain to gender, race, religion, nationality, ethnicity, and social class. We would like you to consider your memberships in these particular groups or categories, and respond to the following statements on the basis of how you feel about these groups and your memberships in them.[4]

After the self-esteem scales were collected, subjects were told that they were each to take the SCAT test, which was said to measure interpersonal and intellectual competencies. Also, the SCAT test was said to be a reliable indicator of individuals' ability to process and integrate information and to make deductive inferences. The test consisted of 10 brief character descriptions of a male and a female, ostensibly drawn from actual clinical observations over several months. Subjects' task was to read each pair of descriptions and decide whether the couple remained together for more than 1 year or broke up before that time.

The tests were ostensibly scored by an assistant. Subjects were reminded of the intergroup exercise at the end of the study, and this expectation of future interaction provided the rationale for giving subjects information about their group's performance on the SCAT test. To manipulate threat to collective but not personal self-esteem, subjects were told that they would be given feedback about their own group's performance on the test to get a sense of their group for this later purpose, but that the experimenter was not allowed to give individual feedback until the end of the experiment. Thus, they would be getting an average score for their particular group (A or B), with their own score excluded from the calculations of the group average. Subjects were randomly assigned to receive either group success feedback (an average group score of 9 out of 10 correct) or group failure feedback (an average group score of 3 out of 10 correct). Each subject received the score with a group result profile on the SCAT test. Those receiving group failure feedback read that their group was somewhat lacking in social sensitivity, was intellectually immature, and had difficulty processing and responding to social and cognitive information. Those receiving group success feedback read that their group was superior in social and intellectual abilities, was mature, and responded well to personal and cognitive challenges.

After receiving group feedback, subjects rated themselves, above-average scorers on the SCAT, below-average scorers on the SCAT, Group A, and Group B (targets were randomly ordered) on 16 adjectives on 7-point scales. Ratings of above- and below-average scorers were made without reference to membership in Group A or Group B, and ratings of Group A and Group B were made without reference to scores on the SCAT. The adjectives concerned both intellectual and social competence. They were *motivated, ambitious, boring, creative, rude, stupid, self-centered, friendly, sincere, insensitive, trustworthy, apathetic, uninformed, considerate, intelligent,* and *incompetent.* When rating Group A and Group B, subjects were explicitly told to exclude themselves from the rating of the group to which they belonged. Because they did not know their own score on the test, ratings of above- and below-average scorers on the SCAT test presumably also excluded the self. Subjects then completed manipulation checks for the group feedback manipulation. These assessed the extent to which they were satisfied with the group's score, wished that their group had done better, and thought their group had done better or worse than average. Additional items were included to determine whether the group feedback manipulation had threatened personal self-esteem. These items assessed how well subjects believed they personally did on the SCAT test, what specific score they thought they had received, and how accurate they believed the SCAT test was as a measure of social and cognitive aptitude. All manipulation check ratings were made on 7-point scales, with the exception of predictions of one's own score, which were made on 10-point scales. A final questionnaire assessed suspicion regarding the experimental procedures. Subjects were asked to indicate what they thought the study was about, whether there was anything odd or unusual about the study, and whether there was any aspect of the study about which they had questions or comments. The data from three subjects were discarded because they indicated suspicion about the feedback on the SCAT test. Subjects were then debriefed, asked not to discuss the study with anyone, and dismissed.

To summarize, the design of the study was 2 (High or Low Self-Esteem) × 2 (Group Success or Failure Feedback on the SCAT Test) × 2 (Target of Rating), with either personal or collective self-esteem as an individual difference variable, and with repeated measures on the last factor. Two sets of target ratings were included in the analyses—above- versus below-average scorers on the SCAT, and ingroup versus outgroup

[3] Analyses including the total collective self-esteem scale and the membership, public, and identity subscales generally yielded main effects similar to those obtained with personal self-esteem, such that highs rated all targets more positively than did lows. The importance to identity subscale occasionally failed to show even this main effect. Neither the total scale nor any of the other subscales showed the predicted three-way interaction between self-esteem, feedback, and target for ratings of above- and below-average scorers. A complete description of results obtained with the other subscales can be obtained from the authors.

[4] The collective self-esteem scale asks subjects to consider their memberships in social groups such as race, gender, religion, and social class because, for college students, these are groups that are ascribed rather than achieved. Memberships in groups that are achieved or chosen have implications for one's personal competence, and hence may be related to both personal and collective self-esteem. Our intention in developing this scale was to create a measure that, as much as possible, assessed collective self-esteem *distinct from* personal self-esteem.

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based on the minimal intergroup paradigm. Self-ratings were also analyzed.

Results

Analyses of the collective self-esteem scale indicated that the four types of statements, as predicted, constituted four clusters or subscales (see Luhtanen & Crocker, 1989, for a discussion of item selection for the scale). The analyses reported here are based on the private collective self-esteem subscale. This subscale assesses subjects' judgments of how positive their social groups are, and comes closest to how Tajfel (1982) defined social identity. For simplicity's sake, we refer to this subscale as a measure of collective self-esteem, but the reader should bear in mind that it is the private subscale (i.e., how one evaluates one's social groups or categories, rather than how others evaluate one's groups or how important these groups are to one's self-concept) to which we are referring.

Self-Esteem

Rosenberg (1965) personal self-esteem scores and private collective self-esteem scores were calculated for the 82 subjects; these scales were correlated in the present sample at $r = .34$ ($p < .01$). Analyses of variance (ANOVAs) with collective self-esteem and feedback as between-subjects factors were performed on all dependent measures, using a median split on private collective self-esteem (median = 23). Additional analyses were performed to determine whether any effects obtained with collective self-esteem could be explained by its moderate correlation with personal self-esteem.

Manipulation Checks

On each of the manipulation checks for threat to collective self-esteem, there was a significant main effect of group feedback. Compared to group success feedback subjects, group failure feedback subjects were less satisfied with their group's score (2.55 vs. 6.62), wished more that their group had done better (4.94 vs. 1.70), and indicated their group's score to be lower (2.33 vs. 6.57; all $Fs > 11.6$, $ps < .002$).

On the checks for threat to personal self-esteem, group failure feedback subjects thought that they themselves would perform worse on the test (3.79 vs. 5.22), and predicted a lower score for themselves (5.09 vs. 7.08 on a 10-point scale) than did group success subjects. Group failure feedback subjects also judged the SCAT test to be less accurate, both as a measure of social sensitivity (2.26 vs. 3.49) and as a measure of intellectual competence (1.74 vs. 2.81; all $Fs > 11.6$, $ps < .002$). There were no significant effects of personal self-esteem, collective self-esteem, or interactions involving any of these variables on any of the manipulation checks, with a single exception—in predicting their own scores, low collective self-esteem subjects were more affected by the group feedback (4.50 vs. 7.43 for failure and success, respectively) than were high collective self-esteem subjects (5.50 vs. 6.63), $F(1, 67) = 8.45$, $p < .01$.

Ratings of the self, above- and below-average scorers on the SCAT test, and the ingroup and outgroup based on a minimal intergroup manipulation were analyzed by reverse-scoring the ratings on negative items, and then summing the 16 ratings, creating a measure with a possible range from 16 to 112, with higher scores indicating more positive evaluations. The items included in the index had high internal consistency, as measured by Cronbach's alpha. Alphas ranged from .79 for the self to .93 for the ingroup based on a minimal intergroup manipulation.

Table 1
Ratings of Above- and Below-Average Scorers as a Function of Collective Self-Esteem and Group Feedback on the SCAT Test

	Collective self-esteem			
	High		Low	
Target	Success	Failure	Success	Failure
Above average	92.50	82.25	83.24	82.80
Below average	65.44	73.00	62.76	62.67

Self-Ratings

A 2 (Collective Self-Esteem) × 2 (Group Feedback) ANOVA on subjects' self-ratings revealed only a significant main effect of collective self-esteem, $F(1, 68) = 16.74$, $p < .0002$, such that high collective self-esteem subjects rated themselves higher ($M = 92.55$) than did low collective self-esteem subjects ($M = 85.0$). The same pattern of results was found when subjects were split on personal self-esteem, with high personal self-esteem subjects rating the self more positively than low personal self-esteem subjects ($M = 91.08$ vs. 86.39), $F(1, 69) = 5.92$, $p < .02$. Group feedback had no effect on subjects' self-ratings ($F < 1.0$, *ns*), suggesting that the collective threat manipulation was successful in that it did not threaten personal self-esteem.

Ratings of Above-Average and Below-Average Scorers

A 2 (Low vs. High Collective Self-Esteem) × 2 (Group Success vs. Failure Feedback) × 2 (Above-Average vs. Below-Average Scoring Target) ANOVA with repeated measures on the last factor was conducted on ratings of above- and below-average scorers. On the basis of previous research in which subjects high or low in trait *personal* self-esteem received *personal* success or failure feedback (Crocker et al., 1987), we predicted that subjects high in collective self-esteem would alter their ratings of above- and below-average scorers as a function of group feedback in a manner that enhanced their ingroup, whereas low collective self-esteem subjects would not. The predicted three-way interaction between collective self-esteem, target, and feedback was significant, $F(1, 68) = 7.08$, $p < .01$. The means for this interaction are presented in Table 1. To investigate this interaction further, separate two-way ANOVAs were performed for high and low collective self-esteem subjects. A significant two-way interaction was found for high collective self-esteem subjects, $F(1, 68) = 14.85$, $p < .01$, indicating that these subjects altered their ratings of below- and above-average scorers as a function of group feedback. High collective self-esteem subjects who received group success feedback rated above-average scorers better than did highs who received group failure feedback ($Ms =$

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92.50 and 82.25, respectively), $F(1, 68) = 9.83$, $p < .01$.[5] They also rated below-average scorers worse than highs whose group failed (65.44 vs. 73.00, respectively), $F(1, 68) = 5.35$, $p < .05$. For low collective self-esteem subjects, the interaction between feedback and target was nonsignificant ($F < 1$), indicating that regardless of whether their own group had failed or succeeded, they were equally likely to rate above-average scorers better than below-average scorers (82.80 vs. 62.67 for failure subjects, and 83.24 vs. 62.76 for success feedback, respectively).

In addition to the predicted three-way interaction, the ANOVA revealed several lower-order effects. There was a significant main effect of target, $F(1, 68) = 137.34$, $p < .0001$, such that above-average scorers were rated higher (84.93) than below-average scorers (66.18), as would be expected if subjects believed the SCAT test measured social and intellectual abilities. This was qualified by a significant interaction between feedback and target, $F(1, 68) = 7.65$, $p < .008$, indicating that group success feedback subjects enhanced above-average scorers (87.87) relative to below-average scorers (64.10) to a greater extent than did group failure feedback subjects (82.53 vs. 67.83, respectively). A significant main effect of collective self-esteem, $F(1, 68) = 9.20$, $p < .004$, indicated that subjects high in collective self-esteem gave higher ratings to *both* above- and below-average scorers (78.30) than did those low in collective self-esteem (72.87).

Given the moderate correlation between personal and collective self-esteem in this sample, it is possible that the three-way interaction between collective self-esteem, group feedback, and target could be explained by differences in personal self-esteem, rather than any unique aspect of collective self-esteem. To investigate this possibility, several additional analyses were conducted. First, the ANOVA was repeated with personal self-esteem as a variable instead of collective self-esteem. In addition to replicating those effects not involving self-esteem obtained in the previous analysis, the analysis yielded the main effect of self-esteem (78.10 vs. 74.20 for high and low personal self-esteem subjects, respectively), $F(1, 69) = 4.41$, $p < .04$, but not the self-esteem by feedback by target interaction ($F < 1.2$, *ns*). No other effects involving personal self-esteem reached statistical significance.

Second, analysis of covariance was performed. The three-way interaction between collective self-esteem, feedback, and target was significant, $F(1, 68) = 7.08$, $p < .01$, when personal self-esteem was used as a covariate. The pattern of the adjusted means was virtually identical to that obtained in the ANOVA reported above. The main effect of collective self-esteem was also significant, $F(1, 67) = 6.54$, $p < .02$, when personal self-esteem was used as a covariate.

Finally, we created a measure of collective self-esteem from which all vestiges of personal self-esteem had been removed, by calculating the partial correlation between personal and collective self-esteem and using the residual scores from this analysis as a measure of collective self-esteem corrected for its association with personal self-esteem.[6] Using a median split on this corrected measure (median = 0.357), an ANOVA was performed with corrected collective self-esteem and feedback as between-subjects variables, and target as a repeated measures variable. The three-way interaction between corrected collective self-esteem, feedback, and target was still significant, $F(1, 78) = 6.08$, $p < .02$. Taken together, these analyses indicate that the interactive effects of collective self-esteem and threats to collective self-esteem on ratings of above- and below-average scorers are due to the unique aspects of collective self-esteem, and not its shared variance with personal self-esteem.

It is also possible that the group feedback threatened personal self-esteem instead of or in addition to collective self-esteem. This possibility seems likely given the effects of the group feedback on subjects' predictions of their own score on the test. To examine this possibility an analysis of covariance (ANCOVA) was performed using subjects' predictions of their own score on the SCAT as a covariate. The three-way interaction between collective self-esteem, feedback, and target was still significant when predicted own score was included as a covariate, $F(1, 68) = 6.48$, $p < .02$.

ANCOVA controls only for main effects of the covariate on the dependent measure—it does not control for the possibility that the effects of threat to collective self-esteem may be mediated by the implied threat to personal self-esteem, which, in turn, *interacts* with trait collective self-esteem. To examine this possibility, we split subjects on the median (6.0) on predicted own score on the SCAT, and performed a 2 (Predicted Own Score) × 2 (Collective Self-Esteem) × 2 (Target) ANOVA with repeated measures on the last factor. We reasoned that if the effects of threat to collective self-esteem on ratings of above- and below-average scorers were mediated by effects on predicted own score, then splitting subjects on predicted own score and substituting this variable for threat to collective self-esteem in the analysis should yield a three-way interaction between predicted own score, trait collective self-esteem, and target. Furthermore, this interaction should be at least as significant as that involving threat to collective self-esteem. The three-way interaction was nonsignificant ($F < 1.0$), supporting our contention that the effects of the group threat are not mediated by its effect on personal threat.

Ingroup and Outgroup Ratings Based on the Minimal Intergroup Manipulation

Ratings of Group A and Group B were coded as ingroup and outgroup ratings based on the group to which each subject was assigned. A 2 (Low vs. High Collective Self-Esteem) × 2 (Success vs. Failure Group Feedback) × 2 (Ingroup vs. Outgroup Target) ANOVA with repeated measures on the third factor revealed that, overall, subjects showed ingroup bias. The signifi-

[5] Throughout the results, simple effects tests were used to explore significant interactions. In these post hoc analyses, the mean-square error for the higher-order interaction was used as the denominator of the *F*-ratio (Winer, 1971).

[6] The corrected collective self-esteem score was calculated by first regressing personal self-esteem on collective self-esteem. This yielded a regression equation with a regression coefficient (B) of .2092 and an intercept of 16.03. Hence, CSE = 16.03 + .2092*PSE, where CSE equals collective self-esteem and PSE equals personal (Rosenberg) self-esteem. The corrected collective self-esteem score consists of the residuals when this relationship is controlled. Hence, the formula for the corrected collective self-esteem (CCSE) score was CCSE = CSE − [16.03 + (.2092*PSE)].

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cant main effect of target, $F(1, 66) = 11.74$, $p < .002$, indicated that the ingroup was rated more positively ($M = 80.13$) than the outgroup ($M = 75.63$). Collective self-esteem did not interact significantly with target ($F < 1.0$, *ns*). There was a significant main effect of feedback, $F(1, 66) = 12.99$, $p < .001$, such that group success feedback subjects gave higher ratings to *both* groups (81.87) than did group failure feedback subjects (74.12).

The two main effects of target and feedback were qualified by a significant interaction between feedback and target, $F(1, 66) = 23.11$, $p < .0001$. Group success feedback subjects rated their ingroup better (86.08) than the outgroup (76.05), $F(1, 66) = 12.43$, $p < .01$, whereas group failure feedback subjects were evenhanded in their ratings of ingroup and outgroup (72.20 vs. 75.15, respectively; $F < 1$). Group success subjects rated the ingroup more positively (86.08) than did group failure subjects (72.20), $F(1, 66) = 66.33$, $p < .0001$. However, ratings of the outgroup were unaffected by the group feedback manipulation (76.05 vs. 75.15; $F < 1.0$).

Because none of these main effects or interactions included self-esteem as a factor, results with a median split on personal self-esteem revealed the same pattern of findings. The only effect of self-esteem was a main effect indicating that high collective self-esteem subjects gave higher ratings for *both* groups (81.99) than did low collective self-esteem subjects (74.00), $F(1, 66) = 13.80$, $p < .001$. When split on personal self-esteem, high personal self-esteem subjects gave higher ratings for *both* groups (80.93) than did low personal self-esteem subjects (75.72), $F(1, 67) = 5.37$, $p < .03$. However, when personal self-esteem was used as a covariate, the main effect of collective self-esteem on ratings of A's and B's was still significant, $F(1, 67) = 6.54$, $p < .02$.

Discussion

Social identity theory (Tajfel, 1982; Tajfel & Turner, 1979, 1986; Turner, 1982) posits that the self-concept has two aspects—personal identity and social identity. Previous research has demonstrated that people vary in the extent to which they have a positive personal identity (i.e., high personal self-esteem), and that those who are high in personal self-esteem are particularly likely to enhance their personal identity in the face of personal threat. We proposed that people also vary in the extent to which they have a positive social identity (i.e., high collective self-esteem), and that those who are high in collective self-esteem are particularly likely to protect their social or collective identity in the face of group or collective threat.

Consistent with our predictions, subjects high in collective self-esteem varied their ratings of above- and below-average scorers as a function of their group's performance feedback. Among the high collective self-esteem subjects, those whose group succeeded rated above-average scorers higher than did those whose group failed; similarly, those whose group failed showed higher ratings of below-average scorers than did those whose group succeeded. Subjects who were low in collective self-esteem did not significantly alter their ratings of above-average and below-average scorers as a function of their own group's performance. These results could not be explained by differences in personal self-esteem.

Several analyses indicated that the effect of the group threat manipulation was not mediated by its effects on personal threat, despite the finding that those who received group failure feedback predicted that their own score would be lower than did those who received group success feedback. Given that subjects knew that they had been arbitrarily assigned to the same group, the effects of group feedback on predictions of personal scores may have been due to an anchoring effect (cf. Tversky & Kahneman, 1974). That is, the group average score on the SCAT may have provided an anchor or starting point from which subjects adjusted, or estimated their personal score.

Implications for Social Identity Theory

This study suggests that many of the predictions of social identity theory may be most applicable to individuals who are high in collective self-esteem. According to social identity theory, there are many ways an individual may attempt to restore a threatened social identity, including leaving the group, reevaluating the group attributes from negative to positive (e.g., "Black is beautiful"), and so on. People who are high in collective self-esteem may be more likely to engage in any or all of these strategies than people who are low in collective self-esteem. Thus, our research has identified an individual difference variable that may add to the predictive power of social identity theory.

These results are consistent with the view that high but not low self-esteem individuals engage in self-enhancing or ingroup-enhancing social comparisons following threat. This conclusion contradicts the predictions of Wills (1981), who argued that those who were low in self-esteem were more likely to engage in self-enhancing downward social comparisons. However, it is consistent with previous research indicating that high self-esteem people engage in a variety of self-enhancing or self-serving strategies, whereas low self-esteem people are characterized by the absence of such self-enhancing strategies (see Taylor & Brown, 1988, for a review).

Direct Versus Indirect Enhancement

One puzzling aspect of both the present results and those of Crocker et al. (1987) concerns the tendency of high self-esteem individuals to engage in indirect but not direct enhancement of the self or the ingroup through ratings of above- versus below-average scorers. In the present study, both high and low collective self-esteem individuals rated their ingroup more negatively following failure feedback than following success feedback, but only high collective self-esteem individuals altered their ratings of above- and below-average scorers in a way that mitigated the implications of this feedback. In the Crocker et al. (1987) study, both high and low personal self-esteem individuals rated the self more negatively following personal failure feedback than following success feedback, but only high personal self-esteem individuals altered their ratings of above- and below-average scorers. Why in both of these studies did high self-esteem individuals engage in this relatively indirect form of self-enhancement?[7]

[7] Our finding that high self-esteem individuals engage in indirect self-enhancement, whereas low self-esteem individuals do not, might seem to contradict recent evidence provided by Brown, Collins, and Schmidt

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One explanation for this pattern of results is that high self-esteem individuals are sensitive to the constraints of reality in their self-enhancing distortions. In the Crocker et al. (1987) study, subjects were given explicit performance feedback about the self. Hence, to distort their self-ratings in a self-enhancing manner would contradict explicit information from the experimenter. However, distorting their ratings of above- and below-average scorers, to soften the implications of the feedback they had received, would not contradict explicit feedback from the experimenter. Similarly, in the present study, distorting ingroup ratings would contradict explicit group feedback from the experimenter, whereas distorting ratings of above- and below-average scorers did not contradict "facts" provided by the experimenter. Thus, the results of these studies suggest one strategy by which high self-esteem individuals negotiate between reality testing and self-enhancement. Rather than distort in ways that contradict known facts, high self-esteem individuals may attempt to alter the meaning or implications of those facts.

Self-Esteem and Minimal Groups

Although ingroup bias was observed in this study, neither collective self-esteem nor personal self-esteem moderated the strength of this effect. Indeed, in a series of studies that have replicated the ingroup bias effect in the minimal intergroup situation, we have failed to find evidence that the magnitude of this effect depends on subjects' level of self-esteem (Crocker & Schwartz, 1985; Crocker et al., 1987). This suggests the possibility that, contrary to Tajfel and Turner's social identity theory, ingroup bias in the minimal intergroup situation is unrelated to esteem needs. However, as the present study demonstrates, collective self-esteem does moderate evaluations of ingroups and outgroups when those groups are defined as above- and below-average scorers on a test, and one's group has succeeded or failed.

This pattern of results raises questions about the boundary conditions for these effects. When will collective or personal self-esteem influence evaluations of ingroup and outgroup members? One possibility is that self-esteem will moderate ingroup bias only when the group boundaries are meaningful or have some evaluative connotations. When membership in minimal groups is assigned arbitrarily, as in this study, group membership may fail to activate concerns with self-enhancement. When membership reflects either on one's personal achievements (such as scoring above- or below-average, or being admitted to an exclusive club) or the group's standing (such as being female in a masculine-oriented culture, or Black in a racist society), then personal or collective self-esteem may moderate evaluations of ingroups and outgroups. Of course, few group memberships are as free of meaning and evaluation as membership in a minimal group, so the role of self-esteem in ingroup bias may be more pervasive than research using the minimal intergroup paradigm suggests.

Alternatively, the failure to find moderating effects of collective self-esteem on ingroup bias may have been due to the desire to be (or appear) realistic about the group feedback provided by the experimenter, as noted previously. If so, then collective self-esteem might moderate ingroup bias even in the minimal intergroup situation, when such explicit feedback about group performance is missing.

Conclusions

In contrast to theories of prejudice that suggest that prejudice and discrimination against outgroups are motivated by personal self-esteem needs (cf. Wills, 1981), we suggest, with social identity theorists, that prejudice may represent an attempt to enhance social identity or collective self-esteem. Thus, rather than indicating an indirect attempt to enhance the self by enhancing the ingroup, prejudice may at least sometimes simply represent an attempt to enhance the ingroup, in the service of collective rather than personal esteem needs.

Consistent with social identity theorists, we believe that collective identity is an important aspect of an individual's self-concept. Despite the long-standing interest of social psychologists in the self-concept in general and self-esteem in particular and the recent burgeoning of research in this area, the role of collective aspects of the self, particularly collective self-esteem, has been largely ignored, with the exceptions of social identity theory and Triandis's research on allocentrism (cf. Triandis, 1989). The present research indicates that trait collective self-esteem and threats to collective self-esteem may have important effects in intergroup phenomena. Furthermore, although collective self-esteem appears to be conceptually and empirically distinct from personal self-esteem, the two domains of self-esteem nonetheless appear to show parallel effects. If this is so, then many well-established findings regarding how individuals high or low in personal self-esteem respond to personal threat may have parallels in how individuals high or low in collective self-esteem respond to collective threat.

(1988) that high self-esteem individuals engage in direct self-enhancement and low self-esteem individuals engage in indirect self-enhancement. However, the methods of our studies differ considerably from theirs, as do the definitions of direct versus indirect self-enhancement. Indeed, our conclusion that high self-esteem individuals will not distort "facts" but will alter the meaning or implications of those facts is compatible with their results.

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